Band-1 Computational Experiment: Ramsey R(5,5) Bound Ledger and R(3,3)=6 Methods Validation

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2026-09-11

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Band-1 Computational Experiment: Ramsey $R(5,5)$ Bound Ledger and $R(3,3)=6$ Methods Validation

Honesty stamp. This note does **not** claim that $R(5,5)$ is determined, and does **not** claim a new bound on $R(5,5)$. It records a literature ledger and an independent small-Ramsey certificate used as methods validation.

Abstract

We execute tonight's Band-1 collapse item on Ramsey numbers. From the published literature (cutoff 2026-09), the standing interval is

$$43 \le R(5,5) \le 46.$$

As methods validation for an independent combinatorial search pipeline on this Linux box, we exhaustively certify the classical exact value $R(3,3) = 6$: there exist triangle-free 2-edge-colorings of K_5 (count = 12), and there are zero triangle-free 2-edge-colorings among all 32768 colorings of K_6 (wall time $\approx 0.0464\,\mathrm{s}$). We state explicitly what would be required for an honest exact determination of $R(5,5)$, and why naive enumeration is impossible.

Problem statement

The diagonal Ramsey number $R(s,s)$ is the least n such that every 2-edge-coloring of the complete graph K_n contains a monochromatic clique K_s.

- **Open target (Band 1.02):** exact value of $R(5,5)$.
- **Finite check (methods validation):** exact value of $R(3,3)$, long known to equal 6, re-verified here by exhaustive search.

Tonight's collapse list item 3 asks for a bound ledger (current lower, current upper, next search experiment). Item 4 offers adjacent finite probes; we chose the Ramsey ledger + small certificate as the most runnable honest probe under a $~$10–20 minute wall budget.

Method

Literature ledger for $R(5,5)$

We compiled the standing published bounds:

Bound	Value	Source
Lower	$R(5,5) \geq 43$	Exoo (1989): explicit $(5,5)$-free coloring of K_{42}
Upper	$R(5,5) \leq 46$	Angeltveit–McKay, arXiv:2409.15709 (2024); *J. Graph Theory* (2026)
Prior upper	≤ 48	Angeltveit–McKay (2018)
Prior upper	≤ 49	McKay–Radziszowski (1995/1997 line)
Conjecture	$= 43$	McKay–Radziszowski

Supporting structural facts used in modern attacks: $R(4,5) = 25$; any graph in $\mathcal{R}(5,5,m)$ has degrees in $[m-25, 24]$; there is a public census of 656 Ramsey$(5,5,42)$ graphs.

Independent certificate for $R(3,3) = 6$

Represent a 2-edge-coloring of K_n as a bit-mask on $\binom{n}{2}$ edges (bit = 1 red, = 0 blue). A coloring is *triangle-free* (in the Ramsey sense) if it has no monochromatic K_3.

1. **Lower bound $R(3,3) > 5$.** Color the edges of the 5-cycle C_5 red and the complementary C_5 (pentagram) blue. Direct check: no monochromatic triangle.
2. **Upper bound $R(3,3) \leq 6$.** Enumerate all $2^{\binom{6}{2}} = 2^{15} = 32768$ colorings of K_6 and verify each contains a monochromatic triangle.
3. **Census.** Report triangle-free coloring counts for $n = 1..6$ as a sanity ledger.

Implementation: pure Python 3 (`verify_r33.py`), no external SAT solver required at this scale.

Experiment design

Parameter	Choice
Probe	A — Ramsey ledger + $R(3,3)$ certificate
Slug	`ramsey_r55_ledger`
Host	Linux box, Python 3.13
Search	Exhaustive bit-mask enumeration for $n \leq 6$
Success criterion	Assert $R(3,3) = 6$; compile $R(5,5)$ bound table; no overclaim
Wall budget	$\ll 1$ minute for the certificate; literature search offline

Code path: `/workspace/experiments/band1/ramsey_r55_ledger/`.

Results (numbers!)

$R(3,3) = 6$ **certificate**

- C5 witness valid: monochromatic triangle = **none**.
- K_5 triangle-free 2-colorings: **12** out of 1024.
- K_6 triangle-free 2-colorings: **0** out of **32768**.
- Dedicated K_6 scan wall time: **0.0464 s** (repeat scan consistent).
- Conclusion: $R(3,3) = 6$ (independent exhaustive re-verification).

Small-n census (triangle-free 2-colorings)

n	edges	colorings $2^{\binom{n}{2}}$	triangle-free	seconds
1	0	1	1	0.0000
2	1	2	2	0.0000
3	3	8	6	0.0000
4	6	64	18	0.0001
5	10	1024	12	0.0013
6	15	32768	0	0.0464

$R(5,5)$ **published interval (ledger)**

Standing bound (2026-09): $43 \leq R(5,5) \leq 46$.

- Lower 43: Exoo 1989.
- Upper 46: Angeltveit–McKay 2024/2026.
- This experiment adds **no** new numeric bound.

Naive scale (why $R(5,5)$ is not a homework loop)

n	edges	raw 2-colorings
42	861	2^861 (~10^259.2)
43	903	2^903 (~10^271.8)
45	990	2^990 (~10^298.0)
46	1035	2^1035 (~10^311.6)

Exact determination therefore requires structured search (degree windows, $\mathcal{R}(4,5,\cdot)$ gluing, LP pruning, isomorphism-free generation, SAT certificates)—not raw $2^{\binom{n}{2}}$ enumeration.

Limitations

- The $R(3,3)$ certificate is classical; its value is methods validation, not a new theorem.
- Counts are over labeled colorings (not isomorphism classes). For $n = 5$, the 12 triangle-free colorings are the two orientations/color-swaps of the pentagon/pentagram family in the labeled universe; we do not claim 12 nonisomorphic Ramsey graphs.
- Literature bounds were compiled from arXiv/journal records available to this run; we did not re-verify Exoo's K_{42} coloring or the Angeltveit–McKay ≤ 46 computation.
- No attempt was made to search for a $(5,5)$-free coloring of K_{43} or to empty $\mathcal{R}(5,5,45)$.

Next experiment

1. **Lower-bound attack:** import the public 656-graph $\mathcal{R}(5,5,42)$ census; attempt one-vertex extensions / SAT for zero-defect K_{43} colorings; record defect spectrum (public near-misses have 2 monochromatic K_5s).
2. **Upper-bound attack:** reproduce a *slice* of the Angeltveit–McKay degree/LP pipeline on a reduced subcase (not the full ≤ 46 proof) as a certified micro-replay.
3. **Adjacent Band-1:** off-diagonal $R(3,k)$ or van der Waerden $W(3,k)$ next open value with SAT-lite.
4. **Alternate tonight item 4:** lonely-runner small-n or cap-set exact size in the next open dimension.

References

1. G. Exoo, "A lower bound for $r(5,5)$," *J. Graph Theory* 13 (1989), 97–98.
2. V. Angeltveit, B. D. McKay, "$R(5,5)$ *le*48," *J. Graph Theory* 89 (2018), 5–13.
3. V. Angeltveit, B. D. McKay, "$R(5,5)$ *le*46," arXiv:2409.15709 (2024); *J. Graph Theory* (2026).
4. B. D. McKay, S. P. Radziszowski, "Subgraph counting identities and Ramsey numbers," *J. Combin. Theory Ser. B* 69 (1997), 193–209.
5. S. P. Radziszowski, "Small Ramsey numbers," *Electron. J. Combin.*, Dynamic Survey DS1 (1994–2024/26 editions).
6. R. L. Graham, B. L. Rothschild, J. H. Spencer, *Ramsey Theory*, Wiley (classical $R(3,3) = 6$).

Artifact paths: code `/workspace/experiments/band1/ramsey_r55_ledger/`; log `/workspace/experiments/logs` results `results.json`.